12 General Renormalization Theory

We saw in the previous chapter that calculations in quantum electrodynamics involving one-loop graphs yield divergent integrals over momentum space, but that these infinities cancel when we express all parameters of the theory in terms of "renormalized" quantities, such as the masses and charges that are actually measured. In 1949 Dyson [1] sketched a proof that this cancellation would take place to all orders in quantum electrodynamics. It was immediately apparent (and will be shown here in Sections 12.1 and 12.2) that Dyson's arguments apply to a larger class of theories with finite numbers of relatively simple interactions, the so-called renormalizable theories, of which quantum electrodynamics is just one simple example.

For some years it was widely thought that any sensible physical theory would have to take the form of a renormalizable quantum field theory. The requirement of renormalizability played a crucial role in the development of the modern "standard model" of weak, electromagnetic, and strong interactions. But as we shall see here, the cancellation of ultraviolet divergences does not really depend on renormalizability; as long as we include every one of the infinite number of interactions allowed by symmetries, the so-called non-renormalizable theories are actually just as renormalizable as renormalizable theories.

It is generally believed today that the realistic theories that we use to describe physics at accessible energies are what are known as "effective field theories." As discussed in Section 12.3, these are low-energy approximations to a more fundamental theory that may not be a field theory at all. Any effective field theory necessarily includes an infinite number of non-renormalizable interactions. Nevertheless, as discussed in Sections 12.3 and 12.4, we expect that at sufficiently low energy all the non-renormalizable interactions in such effective field theories are highly suppressed. Renormalizable theories like quantum electrodynamics and the standard model thus retain their special status in physics, though for reasons that are somewhat different from those that originally motivated the assumption of renormalizability in these theories.

12.1 Degrees of Divergence

Let us consider a very general sort of theory, containing interactions of varying types labelled i. Each interaction may be characterized by the number n_{if} of fields of each type f, and by the number d_i of derivatives acting on these fields.

We will start by calculating the "superficial degree of divergence" D of an arbitrary connected one-particle irreducible Feynman diagram in such a theory. This is the number of factors of momentum in the numerator minus the number in the denominator of the integrand, plus four for every independent four-momentum over which we integrate. The superficial divergence is the actual degree of divergence of the integration over the region of momentum space in which the momenta of all internal lines go to infinity together. That is, if $D > 0$, then the part of the amplitude where all internal momenta go to infinity with a common factor $\kappa \to \infty$ will diverge like

$$\int^{\infty} \kappa^{D-1}\, d\kappa. \tag{12.1.1}$$

In the same sense, an integral with degree of divergence $D = 0$ is logarithmically divergent, and an integral with $D < 0$ is convergent, at least as far as this region of momentum space is concerned. We will come back later to the problem posed by subintegrations that behave worse than the integral over this region.

To calculate D, we will need to know the following about the diagram:

$$\begin{aligned} I_f &\equiv \text{number of internal lines of field type } f, \\ E_f &\equiv \text{number of external lines of field type } f, \\ N_i &\equiv \text{number of vertices of interaction type } i. \end{aligned}$$

We will write the asymptotic behavior of the propagator $\Delta_f(k)$ of a field of type f in the form

$$\Delta_f(k) \sim k^{-2+2s_f}. \tag{12.1.2}$$

Looking back at Chapter 6, we see that $s_f = 0$ for scalar fields, $s_f = \frac{1}{2}$ for Dirac fields, and $s_f = 1$ for massive vector fields. More generally, it can be shown that for massive fields of Lorentz transformation type (A, B), we have $s_f = A + B$. Speaking loosely, we may call s_f the "spin." However, dropping terms that because of gauge invariance have no effect, the effective photon propagator $\eta_{\mu\nu}/k^2$ has $s_f = 0$. A similar result applies to a massive vector field coupled to a conserved current, provided the current does not depend on the vector field. It can also be shown that, in the same sense, the graviton field $g_{\mu\nu}$ has a propagator also with $s_f = 0$.

According to Eq. (12.1.2), the propagators make a total contribution to D equal to

$$\sum_f I_f(2s_f - 2). \tag{12.1.3}$$

Also, the derivatives in each interaction of type i introduce d_i momentum factors into the integrand, yielding a total contribution to D equal to

$$\sum_i N_i d_i. \tag{12.1.4}$$

Finally, we need the total number of independent momentum variables of integration. Each internal line can be labelled with a four-momentum, but these are not all independent; the delta function associated with each vertex imposes a linear relation among these internal momenta, except that one delta function only serves to enforce conservation of the external momenta. Thus, the momentum space integration volume elements contribute to D a term

$$4\left[\sum_f I_f - \left(\sum_i N_i - 1\right)\right], \tag{12.1.5}$$

which, of course, is just four times the number of independent loops in the diagram. Adding the contributions (12.1.3), (12.1.4), and (12.1.5), we find

$$D = \sum_f I_f(2s_f + 2) + \sum_i N_i(d_i - 4) + 4. \tag{12.1.6}$$

Eq. (12.1.6) is not very convenient as it stands, because it gives a value for D that seems to depend on the internal details of the Feynman diagram. Fortunately, it can be simplified by using the topological identities

$$2I_f + E_f = \sum_i N_i n_{if}. \tag{12.1.7}$$

(Each internal line contributes two of the lines attached to vertices, while each external line contributes only one.) Using Eq. (12.1.7) to eliminate I_f, we see that Eq. (12.1.6) becomes

$$D = 4 - \sum_f E_f(s_f + 1) - \sum_i N_i \Delta_i, \tag{12.1.8}$$

where Δ_i is a parameter characterizing interactions of type i:

$$\Delta_i \equiv 4 - d_i - \sum_f n_{if}(s_f + 1). \tag{12.1.9}$$

This result could have been obtained by simple dimensional analysis, without considering the structure of Feynman diagrams. The propagator of a field is a four-dimensional Fourier transform of the vacuum expectation value of a time-ordered product of a pair of free fields, so a conventionally normalized field f whose dimensionality[1] in powers of momentum is $\mathcal{D}_f$ will have a propagator of dimensionality $-4 + 2\mathcal{D}_f$. Hence if the propagator behaves like k^{-2+2s_f} when k is much larger than the mass, then the field must have a dimensionality with $-4 + 2\mathcal{D}_f = -2 + 2s_f$, or $\mathcal{D}_f = 1 + s_f$. An interaction i with n_{if} such fields and d_i derivatives will then have dimensionality $d_i + \sum_f n_{if}(1 + s_f)$. But the action must be dimensionless, so each term in the Lagrangian density must have dimensionality $+4$ to cancel the dimensionality -4 of d^4x. Hence the interaction must have a coupling constant of dimensionality $4 - d_i - \sum_f n_{if}(1 + s_f)$, which is just the parameter Δ_i. The momentum space amplitude corresponding to a connected Feynman graph with E_f external lines of type f is the Fourier transform over $4\sum_f E_f$ coordinates of a vacuum expectation value of the time-ordered product of fields with a total dimensionality $\sum_f E_f(1 + s_f)$, so it has dimensionality $\sum_f E_f(-3 + s_f)$. Of this dimensionality, -4 comes from a momentum space delta function, and $\sum_f E_f(-2 + 2s_f)$ is the dimensionality of the propagators for the external lines, so the momentum space integral itself together with all coupling constant factors has dimensionality

$$\sum_f E_f(-3 + s_f) - (-4) - \sum_f E_f(-2 + 2s_f) = 4 - \sum_f E_f(s_f + 1).$$

The coupling constants for a given Feynman graph have total dimensionality $\sum_i N_i \Delta_i$, leaving the momentum space integral with dimensionality $4 - \sum_f E_f(s_f + 1) - \sum_i N_i \Delta_i$. As long as we are interested in the region of integration where all momenta go to infinity

[1] In this chapter, "dimensionality" will always refer to the dimensionality in powers of mass or momentum, in units with $\hbar = c = 1$. We are using fields that are conventionally normalized, in the sense that the term in the free-field Lagrangian with the largest number of derivatives (which determines the asymptotic behavior of the propagator) has a dimensionless coefficient.

Table 12.1. Terms in the Lagrangian density for quantum electrodynamics. Here d_i, $n_{i\gamma}$, and n_{ie} are the numbers of derivatives, photon fields, and electron fields in the interaction, and Δ_i is the dimensionality of the corresponding coefficient. (Recall that $s_\gamma = 0$, $s_e = \frac{1}{2}$.)

Interaction	d_i	$n_{i\gamma}$	n_{ie}	Δ_i
$-ie\bar{\psi}\not{A}\psi$	0	1	2	$4-1-3=0$
$-\frac{1}{4}(Z_3-1)F_{\mu\nu}F^{\mu\nu}$	2	2	0	$4-2-2=0$
$-(Z_2-1)\bar{\psi}\not{\partial}\psi$	1	0	2	$4-1-3=0$
$[-(Z_2-1)m+Z_2\delta m]\bar{\psi}\psi$	0	0	2	$4-3=1$

together, the degree of divergence of the momentum space integral is its dimensionality, thus justifying Eq. (12.1.8).

If all interactions have $\Delta_i \geq 0$, then Eq. (12.1.8) provides an upper bound on D that depends only on the numbers of external lines of each type, i.e., on the physical process whose amplitude is being calculated:

$$D \leq 4 - \sum_f E_f(s_f+1). \tag{12.1.10}$$

For example, in the simple version of quantum electrodynamics studied in the previous chapter, the Lagrangian included terms of the types shown in Table 12.1. All interactions here have $\Delta_i \geq 0$, and hence a Feynman diagram with E_γ external photon lines and E_e external Dirac lines will

have superficial degree of divergence bounded by Eq. (12.1.10):

$$D \leq 4 - \frac{3}{2}E_e - E_\gamma. \tag{12.1.11}$$

Only a finite number of sets of external lines can yield superficially divergent integrals; these will be enumerated in Section 12.2. We are going to show that the limited number of divergences that appear in theories with $\Delta_i \geq 0$ for all interactions are automatically removed by a redefinition of a finite number of physical constants and a renormalization of fields. For this reason, such theories are called renormalizable. In Section 12.3 we will catalog all the renormalizable theories, and discuss the significance of renormalizability as a criterion for physical theories.

The term “renormalizable” is also applied to individual interactions. Renormalizable interactions are those with $\Delta_i \geq 0$, whose coupling constants have positive or zero dimensionality. Sometimes one distinguishes between interactions with $\Delta_i = 0$, called simply renormalizable, and those with $\Delta_i > 0$, called superrenormalizable. Since adding additional fields or derivatives always lowers Δ_i, there can only be a finite number of renormalizable interactions involving fields of any given types. We have seen that all the interactions in the simplest version of quantum electrodynamics are renormalizable, with the $\bar{\psi}\psi$ terms superrenormalizable.

On the other hand, if any interaction has $\Delta_i < 0$, the degree of divergence (12.1.8) becomes larger and larger the more such vertices we include. No matter how large we take

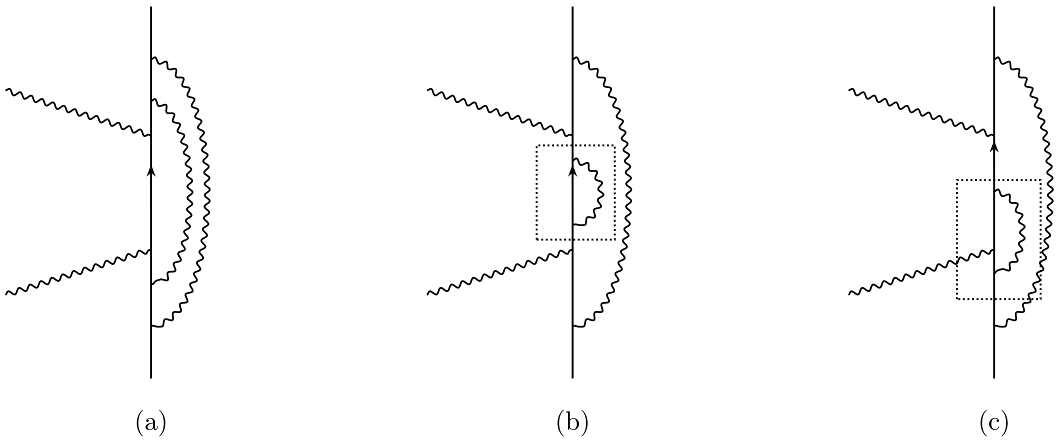

Figure 12.1. Some two-loop graphs for Compton scattering. Here straight lines are electrons; wavy lines are photons. The momentum space integral for diagram (a) is convergent, while for (b) and (c) it is divergent, due to the subintegration associated with the subgraphs surrounded by dotted lines.

the various E_f, eventually with enough vertices of type i for which $\Delta_i < 0$, Eq. (12.1.8) will become positive (or zero), and the integral will diverge. Such interactions, whose couplings have negative-definite dimensionality, are called non-renormalizable;[2] theories with any non-renormalizable interactions are also known as non-renormalizable. But this does not mean that such theories are hopeless; we shall see that these divergences may also be absorbed into a redefinition of the parameters of the theory, but here we need an infinite number of couplings.

It should be kept in mind that we have here calculated the degree of divergence of Feynman diagrams arising only from regions of momentum space in which all internal four-momenta go to infinity together. Divergences can also arise from regions in which only the four-momenta of lines belonging to some subgraph go to infinity. For instance, in quantum electrodynamics Eq. (12.1.11) gives $D \leq -1$ for Compton scattering (where $E_e = 2$, $E_\gamma = 2$), and indeed graphs like Figure 12.1(a) are convergent, but a graph like Figure 12.1(b) or 12.1(c) is logarithmically divergent, because these graphs contain subgraphs (indicated by dotted boxes) with $D \geq 0$. We can think of the divergence of these graphs as being due to an anomalously bad asymptotic behavior that occurs when the eight components of the two independent internal four-momenta of these graphs go to infinity on a particular four-dimensional subspace, namely, that subspace in which the only four-momentum actually going to infinity is the one circulating in the loops that are inserted in the internal lines or at an electron-photon vertex.

It has been shown [2] that the requirement for the actual convergence of

the amplitude corresponding to any graph is that power-counting should give $D < 0$ not only for the complete multiple integral for the whole amplitude, but also for any

[2] In perturbative statistical mechanics, non-renormalizable interactions are called irrelevant, because they become less important in the limit of low energies. Renormalizable and super-renormalizable interactions are called marginal and relevant, respectively.

subintegration defined by holding any one or more linear combinations of the loop momenta fixed. (The graphs shown in Figures 12.1(b) and 12.1(c) fail this test because $D \geq 0$ for the subintegrations in which only the momenta for the loops within the dotted squares are integrated.) We will not repeat the rather long proof here, because it is well treated in earlier books [3], and in any case the method of proof has little to do with how we actually do calculations. The next section will describe how this requirement is fulfilled.

12.2 Cancellation of Divergences

Consider a Feynman diagram, or part of a Feynman diagram, with positive superficial degree of divergence, $D \geq 0$. The part of the momentum space integral where all internal momenta go to infinity together will then diverge, like $\int^{\infty} k^{D-1}\, dk$. If we differentiate $D+1$ times with respect to any external momentum, we lower the net number of momentum factors in the integrand by $D+1$, [3] and hence render this part of the momentum space integral convergent. There may still be divergences arising from subgraphs, like those in Figures 12.1(b) and 12.1(c); for the moment we will ignore this possibility, returning to it later in this section. Since differentiation $D+1$ times renders the integral finite, it follows that the contribution of such a graph or subgraph can be written as a polynomial of order D in external momenta, with divergent coefficients, plus a finite remainder.

To see how this works without irrelevant complications, consider the logarithmically divergent one-dimensional integral

$$\mathcal{J}(q) = \int_0^{\infty} \frac{dk}{k+q}$$

with $D = 1 - 1 = 0$. Differentiating once gives

$$\mathcal{J}'(q) = -\int_0^{\infty} \frac{dk}{(k+q)^2} = -\frac{1}{q},$$

so

$$\mathcal{J}(q) = -\ln q + c.$$

The constant c is obviously divergent, but the rest of the integral is perfectly finite. In exactly the same way, we can evaluate the $D = 1$ integral

$$\int_0^{\infty} \frac{k\, dk}{k+q} = a + bq + q \ln q$$

with divergent constants a and b.

Now, a polynomial term in external momenta is just what would be produced by adding suitable terms to the Lagrangian: if a graph with E_f external lines of type f has degree of divergence $D \geq 0$, then the ultraviolet divergent polynomial is the same as would be produced by adding various interactions i with $n_{if} = E_f$ fields of type f and $d_i \leq D$ derivatives.

[3]For instance, if an internal scalar field line carries a momentum $k + p$, where p is a linear combination of external four-momenta and k is a four-momentum variable of integration, then the derivative of the propagator $[(k+p)^2 + m^2]^{-1}$ with respect to p^μ gives $-2(k_\mu + p_\mu)[(k+p)^2 + m^2]^{-2}$, which goes as k^{-3} rather than k^{-2} for $k \to \infty$.

If there already are such interactions in the Lagrangian, then the ultraviolet divergences simply add corrections to the coupling constants of these interactions. Hence these infinities can be cancelled by including suitable infinite terms in these coupling constants. All that we ever measure is the sum of the bare coupling constant and the corresponding coefficient from one of the divergent polynomials, so if we demand that the sum equals the (presumably finite) measured value, then the bare coupling must automatically contain an infinity that cancels the infinity from the divergent integral over internal momenta. (One qualification: where the divergence occurs in a graph or subgraph with just two external lines, which appears as a radiative correction to a particle propagator, we must demand not that some effective coupling constant equals its measured value, but rather that the complete propagator has a pole at the same position and with the same residue as for free particles.) In this way, all infinities are absorbed into a redefinition of couplings constants, masses, and fields.

For this renormalization program to work, it is essential that the Lagrangian include *all* interactions that correspond to the ultraviolet divergent parts of Feynman amplitudes. (There are exceptions to this rule in supersymmetric theories [4].) The interactions in the Lagrangian are, of course, limited by various symmetry principles, such as Lorentz invariance, gauge invariance, etc., but these constrain the ultraviolet divergences in the same way. (It takes some work to prove that non-Abelian gauge symmetries constrain infinities in the same way that they constrain interactions. This will be shown in Volume II.) In the general case, there are no other limitations on the ultraviolet divergences, so *the Lagrangian must include every possible term consistent with symmetry principles.*

But there is an important class of theories with only a finite number of interactions, where the renormalization program also works. These are the so-called renormalizable theories, whose interactions all have $\Delta_i \geq 0$. Equation (12.1.8) then gives

$$D \leq 4 - \sum_f E_f(s_f + 1),$$

so divergent polynomials arise in only a limited number of Feynman graphs or subgraphs: those with few enough external lines so that $D \geq 0$. The contribution of such divergent polynomials is just the same as would be produced by replacing the divergent graph or subgraph with a single vertex arising from a term in the Lagrangian with E_f fields of type f and $0, 1, \ldots, D$ derivatives. But, comparing with Eq. (12.1.9), we see that *these are precisely the same as the interactions that satisfy the renormalizability requirement* $\Delta_i \geq 0$, or in other words,

$$0 \leq d_i \leq 4 - \sum_f n_{if}(s_f + 1).$$

In order for all infinities to cancel in a renormalizable theory, it is usually necessary that *all* renormalizable interactions that are allowed by symmetries must actually appear in the Lagrangian. [4] For instance, if there is a scalar (or pseudoscalar) field ϕ and fermion field ψ with interactions $\bar{\psi}\psi\phi$ (or $\bar{\psi}\gamma_5\psi\phi$) then we cannot exclude an interaction ϕ^4; otherwise

[4] In addition, interactions and mass terms that are not allowed by global symmetries may appear in the Lagrangian, as long as they are superrenormalizable, that is, with $\Delta_i > 0$. This is because the presence

there would be no counterterm to cancel the logarithmic divergence arising from fermion loops with four attached scalar or pseudoscalar lines.

Let's see in more detail how the cancellation of infinities works in the simplest version of quantum electrodynamics. Equation (12.1.11) shows that the only graphs or subgraphs that could possibly yield divergent integrals are the following:

$E_e = 2,\ E_\gamma = 1$

This is the electron–photon vertex $\Gamma^{*(\ell)}_\mu(p', p)$. (The superscript ℓ indicates that this includes only contributions from graphs with loops.) It has $D = 0$, so its divergent part is momentum-independent. Lorentz invariance then only allows this divergent constant to be proportional to γ_μ, so

$$\Gamma^{*(\ell)}_\mu = L\gamma_\mu + \Gamma^{(f)}_\mu, \tag{12.2.1}$$

with L a logarithmically divergent constant, and $\Gamma^{(f)}_\mu$ finite. This does not uniquely define the constant L, since we can always move a finite term $\delta L\gamma_\mu$ from $\Gamma^{(f)}_\mu$ to $L\gamma_\mu$. To complete the definition, we may note that as shown in Section 9.7, the mass-shell matrix element of $\Gamma^*_\mu(p, p)$ and hence of $\Gamma^{(f)}_\mu(p, p)$ between mass-shell Dirac spinors is proportional to the same matrix element of γ^μ, so we may define L by the prescription that

$$\bar{u}(p, \sigma')\Gamma^{(f)}_\mu(p, p)u(p, \sigma) = 0 \tag{12.2.2}$$

for $p^2 + m_e^2 = 0$.

$E_e = 2,\ E_\gamma = 0$

This is the electron self-energy insertion $\Sigma^*(p)$. It has $D = 1$, so its divergent part is linear in the momentum p^μ carried by the incoming and outgoing fermion. Lorentz invariance (including parity conservation) will only allow it to be a function of $\not{p}$, so we may write the loop contribution as

$$\Sigma^{*(\ell)}(p) = A - (i\not{p} + m)B + \Sigma^{(f)}(\not{p}), \tag{12.2.3}$$

where A and B are divergent constants, and $\Sigma^{(f)}$ is finite. Again, this does not uniquely define the constants A and B, because we can always shift $\Sigma^{(f)}$ by a finite first-order polynomial in $\not{p}$. We will define A and B by the prescription that

$$\Sigma^{(f)} = \frac{\partial \Sigma^{(f)}}{\partial \not{p}} = 0 \quad \text{for} \quad i\not{p} = -m. \tag{12.2.4}$$

Actually, B is not a new divergent constant. As long as we use a regularization procedure that respects current conservation, Γ_μ and Σ will be related by the Ward identity (10.4.27)

$$\Gamma^\mu(p, p) = \gamma^\mu + i\frac{\partial}{\partial p_\mu}\Sigma(p)$$

of a superrenormalizable coupling lowers the degree of divergence, so that the symmetry breaking does not affect those divergences that are cancelled by the strictly renormalizable couplings with $\Delta_i = 0$. Note that it is the *bare* strictly renormalizable couplings that would exhibit the symmetry; renormalized couplings that are defined in terms of mass-shell matrix elements generally show the effect of symmetry breaking.

and therefore

$$L\gamma_\mu + \Gamma^{(f)}_\mu(p,p) = B\gamma_\mu + i\frac{\partial \Sigma^{(f)}(p)}{\partial p^\mu}. \tag{12.2.5}$$

Taking the matrix element of this equation between $\bar{u}(p,\sigma')$ and $u(p,\sigma)$ and using Eqs. (12.2.2) and (12.2.4), we find

$$L = B. \tag{12.2.6}$$

$E_\gamma = 2$, $E_e = 0$

This is the photon self-energy insertion $\Pi^*_{\mu\nu}(q)$. It has $D = 2$, so its divergent part is a second-order polynomial in q. Lorentz invariance only allows $\Pi^*_{\mu\nu}$ to take the form of a linear combination of $\eta_{\mu\nu}$ and $q_\mu q_\nu$ with coefficients depending only on q^2, so the loop contributions take the form

$$\Pi^{*(\ell)}_{\mu\nu}(q) = C_1\eta_{\mu\nu} + C_2\eta_{\mu\nu}q^2 + C_3 q_\mu q_\nu + \text{finite terms},$$

where C_1, C_2, and C_3 are divergent constants. As long as we use a regularization technique that respects current conservation, we must have

$$q^\mu \Pi^{*(\ell)}_{\mu\nu}(q) = 0.$$

The same must then be true for the divergent terms, so $C_1 q_\nu + (C_2 + C_3)q^2 q_\nu$ must be finite for all q. It follows that C_1 and $C_2 + C_3$ must be finite, and can therefore be lumped into the finite part of $\Pi^{*(\ell)}_{\mu\nu}(q)$. Thus

$$\Pi^{*(\ell)}_{\mu\nu}(q) = (\eta_{\mu\nu}q^2 - q_\mu q_\nu)\big(C + \pi(q^2)\big), \tag{12.2.7}$$

where $\pi(q^2)$ is finite and C is the sole remaining divergence in $\Pi_{\mu\nu}$. To pin down the definition of C, we may move any finite constant $\pi(0)$ into C, so that

$$\pi(0) = 0. \tag{12.2.8}$$

$E_\gamma = 4$, $E_e = 0$

This is the amplitude $M_{\mu\nu\rho\sigma}$ for scattering of light by light. It has $D = 0$, so using Lorentz invariance and Bose statistics, it may be written (there is no non-loop contribution)

$$M_{\mu\nu\rho\sigma} = K(\eta_{\mu\nu}\eta_{\rho\sigma} + \eta_{\mu\rho}\eta_{\nu\sigma} + \eta_{\mu\sigma}\eta_{\nu\rho}) + \text{finite terms}$$

with K a potentially divergent constant. However, current conservation gives

$$q^\mu M_{\mu\nu\rho\sigma} = 0$$

and so $K(q_\nu\eta_{\rho\sigma} + q_\rho\eta_{\nu\sigma} + q_\sigma\eta_{\nu\rho})$ is finite. In order for this to be true for $q \neq 0$, K must itself be finite. This is a nice example of the role of symmetry principles in the renormalization program: if K had turned out to be infinite it could not be removed by renormalization of the coupling constant for an interaction $(A_\mu A^\mu)^2$, because no such interaction is allowed by gauge invariance, but K is finite because of current conservation conditions that are imposed by gauge invariance.

$E_\gamma = 1$**,** $E_e = 0$ **and** $E_e = 1$**,** $E_\gamma = 0, 1, 2$

These have $D = 3$ and $D = \frac{5}{2}, \frac{3}{2}$, and $\frac{1}{2}$, respectively, but Lorentz invariance makes all such graphs vanish.

$E_\gamma = 3$**,** $E_e = 0$

This has $D = 1$, but vanishes because of charge-conjugation invariance.

The reader will perhaps have noticed that the independent divergent constants A, B, C are in one-to-one correspondence with the independent parameters Z_2, Z_3, and δm in the counterterm part (11.1.9) of the Lagrangian for quantum electrodynamics. These counterterms make a direct contribution $Z_2 \delta m - (Z_2 - 1)(i\not{p} + m)$ to $\Sigma^*(p)$. The requirement that the position and residue of the one-particle pole be the same as in the free-field propagator means we must choose Z_2 and δm so that the total $\Sigma^*(p)$ satisfies Eq. (12.2.4), i.e.,

$$Z_2 \delta m = -A, \tag{12.2.9}$$

$$Z_2 - 1 = -B, \tag{12.2.10}$$

so that the complete electron self-energy insertion is just the finite function $\Sigma^{(f)}(p)$:

$$\Sigma(p) = \Sigma^{(f)}(p). \tag{12.2.11}$$

Also, $\mathcal{L}_2$ makes a direct contribution to Γ_μ equal to $(Z_2 - 1)\gamma_\mu$. Using Eq. (12.2.6), we see that the full vertex is

$$\Gamma_\mu = \gamma_\mu + (Z_2 - 1)\gamma_\mu + \Gamma_\mu^{*(\ell)} = \gamma_\mu + \Gamma_\mu^{(f)}. \tag{12.2.12}$$

This is not only finite, but satisfies the condition

$$\bar{u}(p, \sigma')\Gamma_\mu(p, p)u(p, \sigma) = \bar{u}(p, \sigma')\gamma_\mu u(p, \sigma), \tag{12.2.13}$$

as can also be seen from Eqs. (10.6.13) and (10.6.14). Finally, $\mathcal{L}_2$ makes a contribution $-(Z_3 - 1)(q^2\eta_{\mu\nu} - q_\mu q_\nu)$ to $\Pi^*_{\mu\nu}(q)$. In order that the photon propagator should have a pole with the same residue as for free fields we need the coefficient of $q^2\eta_{\mu\nu} - q_\mu q_\nu$ in the total $\Pi_{\mu\nu}(q)$ to vanish, so

$$Z_3 = 1 + C \tag{12.2.14}$$

and the photon propagator is then finite:

$$\Pi_{\mu\nu}(q) = (\eta_{\mu\nu}q^2 - q_\mu q_\nu)\pi(q^2). \tag{12.2.15}$$

So far, we have only checked that the divergences, arising from the region of momentum space in which all internal momenta are large (and with generic ratios), are polynomials in external momenta that are cancelled by suitable counterterms. Such graphs are called *superficially convergent*. Before we conclude that all ultraviolet divergences actually are removed by renormalization, we need to consider the ultraviolet divergences arising in higher-order graphs when some subset of the momentum space integration variables rather than all of them go to infinity. For instance, in quantum electrodynamics the superficial divergences in subintegrations come from subgraphs that are either photon self-energy parts Π^*, or electron self-energy parts Σ^*, or electron–electron–photon vertices Γ^μ.

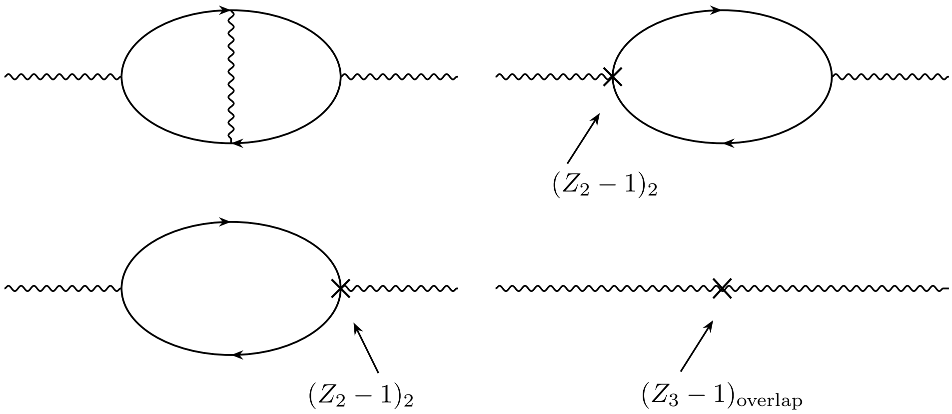

Figure 12.2. Some fourth-order graphs for the photon self-energy in quantum electrodynamics that involve overlapping divergences. Lines carrying arrows are electrons; wavy lines photons. The crosses mark the contribution of counterterms.

The problem with such divergences is that they cannot be removed by differentiation with respect to external momenta; we are left with terms where the derivatives act only on internal lines in the parts of the graph which are *not* in the divergent subgraphs, and therefore do not reduce the degree of divergence of these subgraphs. As mentioned in the previous section, a graph or sum of graphs is actually convergent only if it and all its subintegrations are superficially convergent, in the sense of counting powers of momentum. But wherever such a divergent subgraph appears, it always comes accompanied with an infinite counterterm. In electrodynamics, these are the terms in Eq. (11.1.9): a term $-(Z_3-1)(q^2\eta_{\mu\nu}-q_\mu q_\nu)$ for each $\Pi^*_{\mu\nu}(q)$; a term $Z_2\delta m-(Z_2-1)(i\not{p}+m)$ for each $\Sigma^*(p)$; and a term $(Z_2-1)\gamma^\mu$ for each Γ^μ. Just as for the graph as a whole, these counterterms cancel the infinities from the divergent subgraphs.

Unfortunately, there is a flaw in this simple argument—the possibility of overlapping divergences. That is, it is possible that two divergent subgraphs may share an internal line, so that we cannot regard them as independent divergent integrals. In quantum electrodynamics this happens only when two electron–electron–photon vertices overlap inside a photon or an electron self-energy insertion, [5] as shown in Figures 12.2 and 12.3.

A complete treatment of renormalization that takes account of overlapping divergences should include a prescription for eliminating superficial ultraviolet divergences, not only in the overall integration but in all subintegrations as well, together with a proof that this prescription is (at least formally) implemented by renormalization of masses, fields, and coupling constants. The theorem of Ref. [2] then ensures that all Green's functions of renormalized fields are finite when expressed in terms of renormalized masses and couplings.

[5]The sharing of a line in two self-energy insertions or in a self-energy insertion and a vertex part would not leave enough external lines to attach such a subgraph to the rest of the diagram. Historically, the Ward identity (10.4.26) was used to by-pass the problem of overlapping divergences in the electron self-energy, by expressing the electron self-energy in terms of the vertex function, where overlapping divergences do not occur. This approach will not be followed here, as it is unnecessary, and in any case does not solve the problem for the self-energy of the photon or other neutral particles.

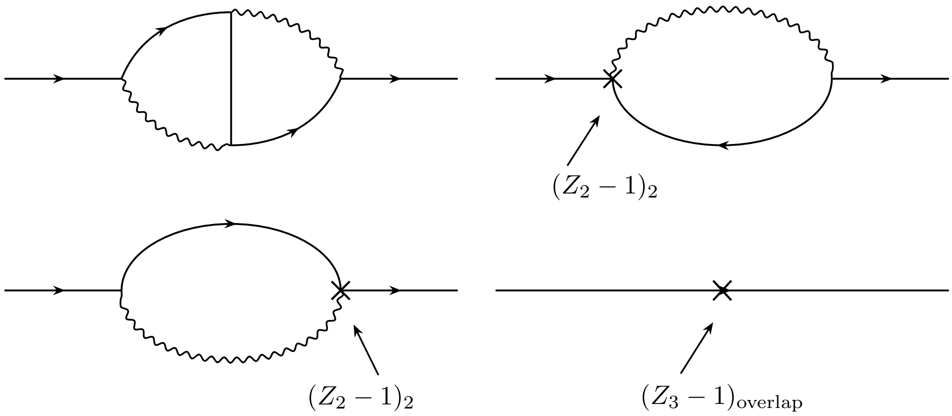

Figure 12.3. Some fourth-order graphs for the electron self-energy in quantum electrodynamics that involve overlapping divergences. Wavy lines are photons, other lines are electrons. The crosses mark the contribution of counterterms.

The first proof that the renormalization of fields, couplings, and masses renders the whole integration and all its subintegrations superficially divergent was offered by Salam [5]. A more specific prescription for eliminating ultraviolet divergences was given by Bogoliubov and Parasiuk [6], and corrected by Hepp [7], and shown by them to be equivalent to a renormalization of fields, masses, and coupling constants. Finally, Zimmermann [8] proved that this prescription does eliminate all superficial divergences in the whole integration and all its subintegrations, and used the theorem of Ref. [2] to conclude from this that the renormalized Feynman momentum-space integrals are convergent.

Briefly, the 'BPHZ' prescription for eliminating superficial divergences requires that we consider all possible ways (called 'forests') of surrounding a whole graph and/or its subgraphs with boxes that may be nested within each other but do not overlap. (An example is given below.) For each forest we define a subtraction term by replacing the integrand for any subgraph of superficial divergence D within a box (starting with the innermost boxes and working outwards) with the first $D + 1$ terms of its Taylor series expansion in the momenta flowing into or out of that box. [6] The subtracted Feynman amplitude is given by the original graph, minus all these subtraction terms, including the subtraction term for a forest consisting of a single box surrounding the whole graph.

It is fairly easy to see that the subtracted Feynman amplitude that is calculated in this way is the same as would be obtained by replacing all fields, coupling constants, and masses in the original Lagrangian by their renormalized counterparts. The difference between this procedure and the sort of renormalization we carried out in Chapter 11 is that the renormalized fields, coupling constants, and masses are defined in terms of amplitudes at a unconventional renormalization point, where all four-momenta vanish. (In this respect, the one-dimensional divergent integrals discussed at the start of this section provide an ele-

[6] As described here, this prescription applies to unrenormalizable as well as renormalizable theories. In renormalizable theories it implies that there is no subtraction unless the box contains one of the limited number of graphs corresponding to renormalizable terms in the Lagrangian.

mentary example of the BPHZ method of separating divergent terms.) But there is nothing special about this renormalization point; once a Feynman amplitude is made convergent by expressing it in terms of these unconventional renormalized quantities, it can be rewritten in terms of conventionally renormalized fields, couplings, and masses without introducing new infinities.

It is not necessary to use the BPHZ subtraction prescription in practice. Replacing fields, masses, and couplings with their renormalized counterparts (defined using any convenient renormalization points) automatically provides counterterms that cancel all infinities. Instead of proving that the BPHZ subtraction prescription really does make all integrals converge, we shall just look at one example that shows how renormalization works, even in the presence of overlapping divergences.

Consider the fourth-order contribution to the photon self-energy insertion $\Pi^*_{\mu\nu}(q)$ shown in Figure 12.2. (The forests here consist of the whole integral over p and p', the subintegration over p alone, and the subintegration over p' alone.) Including the corresponding counterterms for the vertex parts and photon field renormalization, this has the value

$$\begin{aligned}
[\Pi^*_{\mu\nu}(q)]_{\text{overlap}} = & -\frac{e^4}{(2\pi)^8}\int d^4p\int d^4p'\,\frac{1}{(p-p')^2-i\epsilon} \\
& \cdot\operatorname{Tr}\big\{S(p')\gamma_\nu S(p'+q)\gamma^\rho S(p+q)\gamma_\mu S(p)\gamma_\rho\big\} \\
& -2(Z_2-1)_2\frac{ie^2}{(2\pi)^4}\int d^4p\,\operatorname{Tr}\{\gamma_\nu S(p+q)\gamma_\mu S(p)\} \\
& -(Z_3-1)_{\text{overlap}}(q^2\eta_{\mu\nu}-q_\mu q_\nu),
\end{aligned} \tag{12.2.16}$$

where $S(p)\equiv[-i\not{p}+m]/[p^2+m^2-i\epsilon]$; $(Z_2-1)_2$ is the term in Z_2-1 of second order in e; and $(Z_3-1)_{\text{overlap}}$ is a logarithmically divergent constant of fourth order in e that cancels the terms in $[\Pi^*_{\mu\nu}(q)]_{\text{overlap}}$ of second order in q^μ. The factor 2 in the second term arises because there is a renormalization counterterm Z_2-1 for each of the two vertices in the second-order photon self-energy. Note, however, that the first term here can be thought of *either* as the insertion of a vertex correction given by the p'-integral into a photon self-energy given by the p-integral, or as the insertion of a vertex correction given by the p-integral into a photon self-energy given by the p'-integral, but *not* as the insertion of two independent vertex corrections, because there is only one photon propagator.

To see how to handle the infinities in Eq. (12.2.16), note that

$$\big[(Z_2-1)_2+R_2\big]\gamma_\mu=\frac{ie^2}{(2\pi)^4}\int\frac{d^4p'}{p'^2-i\epsilon}\,\gamma_\rho S(p')\gamma_\mu S(p')\gamma^\rho, \tag{12.2.17}$$

where R_2 is a finite remainder. (Lorentz invariance tells us that the integral on the right is proportional to γ_μ. The difference between this integral and $(Z_2-1)_2\gamma_\mu$ equals the complete renormalized electron–electron–photon vertex to second order in e at zero electron and photon momenta, and is therefore finite.) This allows us to rewrite Eq. (12.2.16) in the form

$$
\begin{aligned}
[\Pi^*_{\mu\nu}(q)]_{\rm overlap} = & -\frac{e^4}{(2\pi)^8}\int d^4p\int d^4p'\Bigg[\frac{1}{(p-p')^2-i\epsilon}\,{\rm Tr}\big\{S(p')\gamma_\nu S(p'+q)\gamma^\rho S(p+q)\gamma_\mu S(p)\gamma_\rho\big\} \\
& \quad -\frac{1}{p'^2-i\epsilon}\,{\rm Tr}\big\{S(p')\gamma_\nu S(p')\gamma^\rho S(p+q)\gamma_\mu S(p)\gamma_\rho\big\} \\
& \quad -\frac{1}{p^2-i\epsilon}\,{\rm Tr}\big\{S(p')\gamma_\nu S(p'+q)\gamma^\rho S(p)\gamma_\mu S(p)\gamma_\rho\big\}\Bigg] \\
& -2R_2\frac{ie^2}{(2\pi)^4}\int d^4p\ {\rm Tr}\{\gamma_\nu S(p+q)\gamma_\mu S(p)\} \\
& -(Z_3-1)_{\rm overlap}(q^2\eta_{\mu\nu}-q_\mu q_\nu).
\end{aligned}
\tag{12.2.18}
$$

First consider integration over p' alone. Each of the first two terms is logarithmically divergent, but their difference is finite. The third term is also logarithmically divergent (with a gauge-invariant regulator), but the divergence in this term (unlike the first two terms) takes the form of a second-order polynomial in q, with the remainder finite. This remaining divergence is cancelled by the term $-(Z_3-1)(q^2\eta^{\mu\nu}-q^\mu q^\nu)$ that cancels all second-order terms in the expansion of $\Pi^*_{\mu\nu}(q)$. So the p'-subintegration gives a finite result. The symmetry of Eq. (12.2.16) shows that in exactly the same way the p subintegration also gives a finite result. Generic subintegrations over p and p' with $ap+bp'$ held fixed (where a and b are arbitrary non-zero constants) are manifestly convergent, and the integration over p and p' together is made finite by the counterterm $-(Z_3-1)(q^2\eta^{\mu\nu}-q^\mu q^\nu)$. Thus Eq. (12.2.18) and any of its subintegrations satisfy the power-counting requirements for convergence, and therefore according to the theorem [2] quoted in the previous section, the whole expression actually converges.

* * *

In electrodynamics there is a natural definition of renormalized couplings as well as renormalized masses and fields. This is not always the case. For example, consider the theory of a single real scalar field $\phi(x)$ with Lagrangian density

$$
\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{1}{24}g\phi^4. \tag{12.2.19}
$$

To one-loop order, the S-matrix for scalar–scalar scattering is given by the Feynman rules as

$$
S(q_1q_2\to q_1'q_2') = \frac{-i(2\pi)^4\delta^4(p_1'+p_2'-p_1-p_2)}{(2\pi)^6(16E_1'E_2'E_1E_2)^{1/2}}F(q_1q_2\to q_1'q_2'), \tag{12.2.20}
$$

where

$$-i(2\pi)^4 F(q_1 q_2 \to q_1' q_2') = \; -i(2\pi)^4 g + \frac{1}{2}\big[-i(2\pi)^4 g\big]^2 \left[\frac{-i}{(2\pi)^4}\right]^2$$
$$\cdot \int d^4k \Bigg[\frac{1}{[(q_1+k)^2+m^2-i\epsilon][(q_2-k)^2+m^2-i\epsilon]}$$
$$+ (q_2 \to -q_1') + (q_2 \to -q_2') \Bigg], \tag{12.2.21}$$

and q_1, q_2 and q_1', q_2' are the incoming and outgoing four-momenta. Combining denominators and rotating the k^0-integration contour as usual, this is

$$F = g - \frac{g^2}{16\pi^2}\int_0^\infty k^3\, dk \int_0^1 dx \Big\{ [k^2+m^2-sx(1-x)]^{-2}$$
$$+ [k^2+m^2-tx(1-x)]^{-2} + [k^2+m^2-ux(1-x)]^{-2} \Big\}, \tag{12.2.22}$$

where $s, t,$ and u are the Mandelstam variables

$$s = -(p_1+p_2)^2, \qquad t = -(p_1-p_1')^2, \qquad u = -(p_1-p_2')^2, \tag{12.2.23}$$

related by $s+t+u = 4m^2$; also, x is the Feynman parameter introduced in combining denominators. With an ultraviolet cutoff at $k = \Lambda$, this gives the result (for $\Lambda \gg m$)

$$F = g - \frac{g^2}{32\pi^2}\int_0^1 dx \Bigg\{ \ln\!\left(\frac{\Lambda^2}{m^2 - sx(1-x)}\right) + \ln\!\left(\frac{\Lambda^2}{m^2 - tx(1-x)}\right)$$
$$+ \ln\!\left(\frac{\Lambda^2}{m^2 - ux(1-x)}\right) - 3 \Bigg\}. \tag{12.2.24}$$

We can define the renormalized coupling g_R as the value of F at any point s, t, u we like, provided we stay in the region where F is real. For instance, suppose that in order to maintain the symmetry among the scalars, we choose to renormalize at the off-mass-shell point [7] $p_1^2 = p_2^2 = p_1'^2 = p_2'^2 = \mu^2$, $s = t = u = -4\mu^2/3$. Defining the renormalized coupling g_R as the value of F at this point, we have

$$g = g_R + \frac{3g_R^2}{32\pi^2}\left[\ln\!\left(\frac{\Lambda^2}{\mu^2}\right) - 1 - \int_0^1 dx\, \ln\!\left(\frac{4x(1-x)}{3} + \frac{m^2}{\mu^2}\right)\right] + \cdots . \tag{12.2.25}$$

The cutoff dependence then cancels in Eq. (12.2.24) to order g_R^2, leaving a finite formula for F in terms of g_R:

$$F = g_R - \frac{g_R^2}{32\pi^2}\int_0^1 dx \Bigg\{ \ln\!\left(\frac{m^2 + 4x(1-x)\mu^2/3}{m^2 - sx(1-x)}\right) + \ln\!\left(\frac{m^2 + 4x(1-x)\mu^2/3}{m^2 - tx(1-x)}\right)$$
$$+ \ln\!\left(\frac{m^2 + 4x(1-x)\mu^2/3}{m^2 - ux(1-x)}\right) \Bigg\} + \cdots . \tag{12.2.26}$$

[7] Going back over the derivation of Eq. (12.2.25), one may check that in this derivation we have not used the conditions $p_1^2 = p_2^2 = p_1'^2 = p_2'^2 = -m^2$, so Eq. (12.2.24) is valid whatever we take for the external line masses.

Here μ^2 may be taken to be any real quantity greater than $-3m^2$, in which range g_R is real. The explicit μ-dependence in Eq. (12.2.26) is, of course, cancelled by the μ-dependence of the renormalized coupling. This freedom to change the renormalization prescription (which of course exists also in electrodynamics and other realistic theories) will be of great importance to us when we come to the renormalization group method in Volume II.

12.3 Is Renormalizability Necessary?

In the previous section we found a special class of theories having only a finite number of terms in the Lagrangian, to which the renormalization program is nevertheless applicable. These are theories in which all interactions satisfy the renormalizability condition

$$\Delta_i \equiv 4 - d_i - \sum_f n_{if}(s_f + 1) \geq 0,$$

where d_i and n_{if} are the numbers of derivatives and fields of type f in interactions of type i, and s_f is (with some qualifications) the spin of fields of type f. For renormalization to work in such theories, it is also usually necessary that all renormalizable interactions that are allowed by symmetry principles should actually appear in the Lagrangian.

It is important that there are only a limited number of such interaction types. Δ_i becomes negative if we have too many fields or derivatives, or fields of too high spin. Barring special cancellations, there are no renormalizable interactions at all involving fields with $s_f \geq 1$, because the only possible term in the Lagrangian with $\Delta_i \geq 0$ that involves such a field along with two or more other fields would involve a single $s_f = 1$ field along with two scalars and no derivatives, which would not be Lorentz-invariant. We shall see in Volume II that general, massless, spin one gauge fields in a suitable gauge effectively have $s_f = 0$, like the photon. Also, in Volume II we shall see that even massive gauge fields may effectively have $s_f = 0$, depending on where their mass comes from. Leaving aside these special cases, Table 12.2 gives a list of *all* renormalizable terms in the Lagrangian density that are allowed by Lorentz invariance and gauge invariance involving scalars ($s = 0$), photons ($s = 0$), and spin $\frac{1}{2}$ fermions ($s = \frac{1}{2}$).

We see that the requirement of renormalizability puts severe restrictions on the variety of physical theories that we may consider. Such restrictions provide a valuable key to the structure of physical theories. For instance, Lorentz and gauge invariance by themselves would allow the introduction of a 'Pauli' term proportional to $\bar{\psi}[\gamma_\mu, \gamma_\nu]\psi F^{\mu\nu}$ in the Lagrangian of quantum electrodynamics, which would make the magnetic moment of the electron an adjustable parameter, but we exclude such terms because they are not renormalizable. The successful predictions of quantum electrodynamics, such as the calculation of the magnetic moment of the electron outlined in Section 11.3, may be regarded as validations of the principle of renormalizability. The same applies to the standard model of weak, electromagnetic, and strong interactions, to be discussed in Volume II; there are any number of terms that might be added to this theory, such as four-fermion interactions among quarks and leptons, that would invalidate all the predictions of the standard model, and are excluded only because they are non-renormalizable.

Table 12.2. Allowed renormalizable terms in a Lagrangian density involving scalars ϕ, Dirac fields ψ, and photon fields A^μ. Here n_{if} and d_i are the number of fields of type f and the number of derivatives in an interaction of type i, and Δ_i is the dimensionality of the associated coefficient.

	n_{if}		d_i	Δ_i	$\mathcal{H}_i$
Scalars	Photons	Spin $\frac{1}{2}$			
1	0	0	0	3	ϕ
2	0	0	0	2	ϕ^2
2	0	0	2	0	$\partial_\mu\phi\partial^\mu\phi$
3	0	0	0	1	ϕ^3
4	0	0	0	0	ϕ^4
2	1	0	1	0	$\phi\partial_\mu\phi A^\mu$
2	2	0	0	0	$\phi^2 A_\mu A^\mu$
1	0	2	0	0	$\phi\bar{\psi}\psi$
0	2	0	2	0	$F_{\mu\nu}F^{\mu\nu}$
0	0	2	0	1	$\bar{\psi}\psi$
0	0	2	1	0	$\bar{\psi}\gamma^\mu\partial_\mu\psi$
0	1	2	0	0	$\bar{\psi}\gamma^\mu A_\mu\psi$

Must we believe that the Lagrangian is restricted to contain only renormalizable interactions? As we saw in the previous section, if we include in the Lagrangian *all* of the infinite number of interactions allowed by symmetries, then there will be a counterterm available to cancel every ultraviolet divergence. In this sense, as said earlier, non-renormalizable theories are just as renormalizable as renormalizable theories, as long as we include all possible terms in the Lagrangian.

In recent years it has become increasingly apparent that renormalizability is not a fundamental physical requirement, and that in fact any realistic quantum field theory will contain non-renormalizable as well as renormalizable terms. This change in point of view can be traced in part to the continued failure to find a renormalizable theory of gravitation. In the general class of metric theories of gravitation governed by Einstein's principle of equivalence there are no renormalizable interactions at all—generally covariant interactions must be constructed from the curvature tensor and its generally covariant derivatives, and hence, even in a 'gauge' where the graviton propagator goes as k^{-2}, these interactions involve too many derivatives of the metric for renormalizability. In particular, we can easily see that general relativity is non-renormalizable from the fact that its coupling constant $8\pi G_N = (2.43 \cdot 10^{18}\ \text{GeV})^{-2}$ has negative dimensionality. Even if nothing else did, the cancellation of divergences due to virtual gravitons would require that the Lagrangian contain all interactions allowed by symmetries—not only interactions involving gravitons, but involving any particles.

But if renormalizability is not a fundamental physical principle, then how do we explain the success of renormalizable theories like quantum electrodynamics and the standard model? The answer can be seen by simple dimensional analysis. We have already noted that the coupling constant of an interaction of type i has dimensionality

$$[g_i] \sim [\text{mass}]^{\Delta_i}, \tag{12.3.1}$$

where Δ_i is the index (12.1.9). Non-renormalizable interactions are just those whose coupling constants have the dimensionality of *negative* powers of mass. Now, it is not unreasonable to guess from (12.3.1) that the coupling constants not only have dimensionalities governed by Δ_i, but are roughly of order

$$g_i \approx M^{\Delta_i}, \tag{12.3.2}$$

where M is some common mass. (This is found to be actually the case in the effective field theories discussed below and in more detail in Volume II.) In calculating physical processes at a characteristic momentum scale $k \ll M$, the inclusion of a non-renormalizable interaction of type i with $\Delta_i < 0$ will introduce a factor $g_i \approx M^{\Delta_i}$, which on dimensional grounds must be accompanied by a factor $k^{-\Delta_i}$, and so the effect of such an interaction is suppressed[8] for $k \ll M$ by a factor $(k/M)^{-\Delta_i} \ll 1$. (This argument will be made more carefully using the method of the renormalization group in Volume II.) The success of the renormalizable theories of electroweak and strong energies shows only that M is very much larger than the energy scale at which these theories have been tested.

For instance, the leading non-renormalizable corrections to the conventional electrodynamics of electrons or muons would be those interactions of dimension 5, which are suppressed by only one factor of $1/M$. There is just one such interaction allowed by Lorentz, gauge, and CP invariance, a Pauli term of order $(ie/2M)\bar{\psi}[\gamma_\mu, \gamma_\nu]\psi F^{\mu\nu}$. According to Eqs. (10.6.24), (10.6.17), and (10.6.19), such a term would contribute an amount of order $4e/M$ to the magnetic moment of the electron or muon. The calculated value of the magnetic moment of the electron agrees with experiment to within terms of order $10^{-10}e/2m_e$, so M must be greater than about $8 \cdot 10^{10} m_e = 4 \cdot 10^7$ GeV.

This limit may be weakened if other symmetries restrict the form of the non-renormalizable interactions. For instance, the conventional Lagrangian of quantum electrodynamics is invariant under a chiral transformation $\psi \to \gamma_5\psi$, except for a change of sign of the fermion mass term $-m\bar{\psi}\psi$. If we assume that the full Lagrangian is invariant under a formal symmetry $\psi \to \gamma_5\psi$, $m \to -m$, then a Pauli term in the Lagrangian would have to appear with an extra factor m/M, so that its contribution to the magnetic moment would be only of order $4em/M^2$. Because of the extra factor of m, here it is the muon rather than

[8] It is essential at this point to assume that the ultraviolet divergences have been removed by renormalization, so that there are no factors of an ultraviolet cutoff Λ to mess up our dimensional analysis. Otherwise, dimensional analysis tells us that for $\Lambda \to \infty$, each additional non-renormalizable coupling constant factor g_i with $\Delta_i < 0$ would be accompanied with a growing factor $\Lambda^{-\Delta_i}$. This dimensional argument led Heisenberg [9] very early to classify interactions according to the dimensionality of their coupling constants, and to suggest [10] that new effects might arise at energies of order g_i^{1/Δ_i}, as for instance at the energy $G_F^{-1/2} \approx 300$ GeV, where G_F is the four-fermion coupling constant of the Fermi beta decay theory. After the development of renormalization theory it was noted by Sakata *et al.* [11] that the non-renormalizable theories are those whose coupling constants have negative dimensionality.

the electron that provides the most useful limit on M. The calculated value of the magnetic moment of the muon agrees with experiment to within terms of order $10^{-8}e/2m_e$, so M must be greater than about $\sqrt{8 \cdot 10^8}\, m_\mu = 3 \cdot 10^3$ GeV. In any case, if M is anywhere near as large as 10^{18} GeV, then we are certainly justified in neglecting any non-renormalizable interactions that might appear in quantum electrodynamics.

These considerations help us to cope with some of the puzzles associated with higher-derivative terms in the Lagrangian. For instance, in the general theory of a real scalar field ϕ, we would expect to find terms in the Lagrangian density of the form $\phi \Box^n \phi$. Any one such term would make a direct contribution to the scalar self-energy function $\Pi^*(q^2)$ proportional to $(q^2)^n$. If we were to include this contribution to all orders, but ignore all other effects of non-renormalizable interactions, then the propagator $\Delta'(q^2) = 1/(q^2 + m^2 - \Pi^*(q^2))$ would not have the simple pole in q^2 at negative q^2 expected from the general arguments of Section 10.7, but n such poles (some of which may coincide), generally at complex values of q^2. But if the non-renormalizable term $\phi \Box^n \phi$ has a coefficient of order $M^{-2(n-1)}$, where $M \gg m$, then the extra poles are at q^2 of order M^2, where it is illegitimate to ignore the infinite number of other non-renormalizable interactions that must also appear in the Lagrangian. Thus the appearance of higher-derivative terms in a general non-renormalizable Lagrangian is not in conflict with the general principles underlying quantum field theory that were used in Section 10.7. But by the same token, we also cannot use higher-derivative terms to avoid ultraviolet divergences altogether, as has been repeatedly proposed. A term $M^{-2(n-1)}\phi \Box^n \phi$ in the Lagrangian density provides a cutoff at momenta $q^2 \approx M^2$, but at these momenta we cannot ignore all the other non-renormalizable interactions that must be present.

Although highly suppressed, non-renormalizable interactions may be detectable if they have effects that would otherwise be forbidden. For instance, we will see in Section 12.5 that the symmetries of charge-conjugation and space-inversion invariance are an automatic consequence of the structure of the electromagnetic interactions that is imposed by gauge invariance, Lorentz invariance, and renormalizability, but we can easily imagine non-renormalizable terms that would violate these symmetries, such as an electron electric dipole moment term $\bar{\psi}\gamma_5[\gamma_\mu, \gamma_\nu]\psi F^{\mu\nu}$, or the Fermi interaction $\bar{\psi}\gamma_5\gamma_\mu\psi\bar{\psi}\gamma^\mu\psi$. It is widely believed today that the conservation of baryon and lepton number is violated by very small effects of highly suppressed non-renormalizable interactions. Another example of a detectable non-renormalizable interaction is provided by gravitation. As mentioned before, gravitons have no renormalizable interactions at all. But, of course, we detect gravitation, because it has the special property that the gravitational fields of all the particles in a macroscopic body add up coherently.

Although non-renormalizable theories involve an infinite number of free parameters, they retain considerable predictive power: [12] they allow us to calculate the non-analytic parts of Feynman amplitudes, like the $\ln q$ and $q \ln q$ terms in the one-dimensional examples at the beginning of the previous section. Such calculations just reproduce the results required by the axiom of S-matrix theory, that the S-matrix has only those singularities required by unitarity.

Paradoxically, it is just in the case where symmetry principles forbid renormalizable interactions that non-renormalizable quantum field theories prove the most useful. In such cases we can derive a useful perturbation theory by expanding in powers of k/M. This has been worked out in detail for the theory of low-energy pions [12,13], to be discussed in detail in Volume II, and the theory of low-energy gravitons [14]. For a simpler example, consider the theory of a real scalar field, satisfying the principle of invariance under the field translation

$$\phi(x) \longrightarrow \phi(x) + \epsilon$$

with ϵ an arbitrary constant. This symmetry forbids any renormalizable interactions or scalar mass, but it allows an infinite number of non-renormalizable derivative interactions

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{g}{4}(\partial_\mu\phi\partial^\mu\phi)^2 - \cdots ;$$

where $g \approx M^{-4}$, and '$\cdots$' denotes terms with more derivatives or fields. (For simplicity, it is assumed here that the theory also has a symmetry under the reflection $\phi \to -\phi$.) According to the above dimensional analysis, the graph for a general reaction in which all energies and momenta are of order $k \ll M$ is suppressed by a factor $(k/M)^\nu$, where

$$\nu = -\sum_i V_i\Delta_i = \sum_i V_i(d_i + n_i - 4),$$

with n_i and d_i the numbers of scalar fields and derivatives in an interaction of type i, and V_i the number of vertices for these interactions in our graph. For $k \ll M$, the dominant contributions to any process are those with the smallest value of ν. The formula for ν can be put in a more useful form by using the familiar topological identities for a connected graph:

$$\sum_i V_i = I - L + 1, \qquad \sum_i V_i n_i = 2I + E,$$

where I, E, and L are the numbers of internal lines, external lines, and loops in our graph. Combining these relations gives

$$\nu = 2E - 4 + 4L + \sum_i V_i(d_i - n_i),$$

Now, the field translation symmetry requires that every field must be accompanied with at least one derivative, so the quantity $d_i - n_i$ as well as L is non-negative for all interactions. Thus for a given process (that is, a fixed number E of external lines) the dominant terms will be those constructed solely from *tree* graphs (i.e., $L = 0$), and interactions with the minimum number $d_i = n_i$ of derivatives. That is, in leading order we can take the Lagrangian density to depend only on *first* derivatives of the field. Higher-order corrections may involve loops and/or interactions with more derivatives on some fields. But to any given order ν in k/M, we need only consider a finite number of graphs, those with $L \leq (4 - 2E + \nu)/4$, and only a finite number of interaction types.

For instance, scalar–scalar scattering is given in leading order by the one-vertex tree graph calculated using the interaction $-g(\partial_\mu\phi\partial^\mu\phi)^2$ in first order. According to our formula for ν, the leading correction, suppressed at low energy by a factor $(k/M)^2$, arises from another single-vertex tree graph, produced by an interaction with two additional derivatives of the form[9] $\partial_\mu \partial_\nu \phi \partial^\mu \partial^\nu \phi \partial_\rho \phi \partial^\rho \phi$. The next corrections, suppressed at low energy by two further factors of k/M, arise both from the one-loop diagram of Figure 12.4 (including permutations of external lines), calculated using only the interaction $-g(\partial_\mu \phi \partial^\mu \phi)^2$, and also from tree graphs with a single vertex arising from a quartic interaction with eight derivatives, whose couplings contain infinite parts that cancel the ultraviolet divergence from the loop graph.[10] The loop graph also yields finite terms in the scattering amplitude proportional to terms like $s^4 \ln s + t^4 \ln t + u^4 \ln u$, $s^2 t^2 \ln u + t^2 u^2 \ln s + u^2 s^2 \ln t$, etc., with calculable coefficients proportional to g^2. These finite terms simply represent the correction to the lowest-order scattering amplitude needed to ensure the unitarity of the S-matrix, but perturbative quantum field theory is by far the easiest way of calculating them.

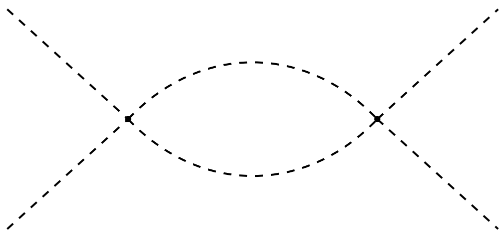

Figure 12.4. One-loop diagram for scalar–scalar scattering in the theory with derivative quadrilinear interactions.

Although non-renormalizable theories can provide useful expansions in powers of energy, they inevitably lose all predictive power at energies of the order of the common mass scale M that characterizes the various couplings. If we were to take these expansions literally, the results for S-matrix elements would violate unitarity bounds for $E \gg M$. There seem to be just two possibilities about what happens at such energies. One is that the growing strength of the effects of the non-renormalizable couplings somehow saturates, avoiding any conflict with unitarity [15]. The other is that new physics of some sort enters at the scale M. In this case, the non-renormalizable theories that describe nature at energies $E \ll M$ are just *effective field theories* rather than truly fundamental theories.

Probably the earliest example of an effective field theory was derived in the 1930s by Euler *et al.* [16], as a theory of low-energy photon–photon interactions. (See Section 1.3.) In effect, they calculated the contribution to photon–photon scattering of Feynman diagrams such as Figure 12.5, and found that at energies much less than m_e the scattering of light by light was the same as would be calculated with an effective Lagrangian

$$\mathcal{L}_{\text{eff}} = \frac{2\alpha^2}{45 m_e^4}\left[(\mathbf{E}^2 - \mathbf{B}^2)^2 + 7(\mathbf{E}\cdot\mathbf{B})^2\right] + \text{higher orders in } \frac{eE}{m_e^2} \;\&\; \frac{eB}{m_e^2}$$

[9] In accordance with the remarks of Section 7.7, we are excluding interactions involving $\Box\phi$, because the field equation for ϕ can be used to express such interactions in terms of the others.

[10] These are the only ultraviolet divergences encountered in one-loop graphs if we use dimensional regularization. For other methods of regularization there are also quartic and quadratic divergences, which are cancelled by counterterms in the four-scalar interactions with four or six derivatives.

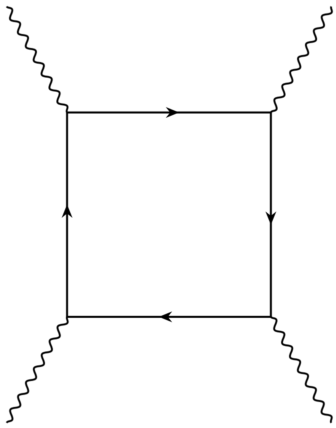

Figure 12.5. Diagram for photon–photon scattering, whose effect at low energy can be calculated from the effective Lagrangian of Euler *et al.* [16]. Straight lines are electrons; wavy lines are photons.

Euler *et al.* used this effective Lagrangian only in the tree approximation, to calculate the leading terms in photon interaction matrix elements. It was not until much later that such Lagrangians, though non-renormalizable, were used beyond the tree approximation [12,17].

In modern jargon, we say that in deriving this Lagrangian the electron is 'integrated out', because in the one-loop approximation we have

$$\exp\left(i\int \mathcal{L}_{\rm eff}(\mathbf{E},\mathbf{B})\,d^4x\right) = \int \left[\prod_x d\psi_e(x)\right] \exp\left(i\int \mathcal{L}_{\rm QED}(\psi_e, A)\,d^4x\right).$$

A more general procedure is simply to write down the most general non-renormalizable effective Lagrangian, use it to calculate various amplitudes as an expansion in energies and momenta, and then choose the constants in the effective Lagrangian by matching the results it gives for these amplitudes to those derived from the underlying theory.

We will encounter effective field theories again, especially in considering broken symmetries in Volume II. As we shall see, effective field theories are useful even where they cannot be derived from an underlying theory, either because the theory is unknown, or because its interactions are too strong to allow the use of perturbation theory. Indeed, even if we knew nothing about the properties of charged particles, the scattering of photons at sufficiently low energy would have to be described by an effective Lagrangian consisting of the terms $(\mathbf{E}^2 - \mathbf{B}^2)^2$ and $(\mathbf{E}\cdot\mathbf{B})^2$, because these are the unique quartic Lorentz- and gauge-invariant terms with no derivatives acting on $\mathbf{E}$ and $\mathbf{B}$. Terms with such derivatives would be suppressed at low photon energies E by additional factors of E/M, where M is some typical mass of the charged particles that are being integrated out. We can go further: we shall see that effective field theories are useful even where the light particles they describe are not present in the underlying theory at all, but composites of the heavy particles that are integrated out. The underlying theory might not even be a field theory at all—the problem of incorporating gravitation has led many theorists to believe that, in fact, it is a string theory. But wherever an effective field theory comes from, it is inevitably a non-renormalizable theory.

12.4 The Floating Cutoff[11]

Before closing this chapter, it is worth commenting on the relation between conventional renormalization theory and an approach pioneered by Wilson [18]. In Wilson's method one imposes a "floating" finite ultraviolet cutoff (either sharp or smooth) at momenta with components of order Λ, and instead of taking $\Lambda \to \infty$, one requires that the bare constants of the theory (those appearing in the Lagrangian) depend on Λ in such a way that all observable quantities are Λ-independent.

It is convenient to work with dimensionless parameters. If a bare coupling or mass parameter $g_i(\Lambda)$ has dimensionality $[\text{mass}]^{\Delta_i}$, we define the corresponding dimensionless parameter $\mathcal{G}_i$ by

$$\mathcal{G}_i(\Lambda) \equiv \Lambda^{-\Delta_i} g_i(\Lambda). \tag{12.4.1}$$

Ordinary dimensional analysis tells us that the value of $\mathcal{G}_i$ at one value Λ' of the cutoff can be expressed as a function of the values of the $\mathcal{G}_j$ at another value Λ of the cutoff, and the dimensionless ratio Λ'/Λ:

$$\mathcal{G}_i(\Lambda') = F_i\big(\mathcal{G}(\Lambda), \Lambda'/\Lambda\big). \tag{12.4.2}$$

No dimensional parameters other than Λ' and Λ can appear in F, because no ultraviolet or infrared divergences can enter here; the difference between the constants at Λ and at Λ' arises from diagrams whose internal lines are restricted to have momenta between Λ and Λ'. Differentiating Eq. (12.4.2) with respect to Λ' and then setting Λ' equal to Λ yields a differential equation for $\mathcal{G}_i$:

$$\Lambda \frac{d}{d\Lambda} \mathcal{G}_i(\Lambda) = \beta_i\big(\mathcal{G}(\Lambda)\big), \tag{12.4.3}$$

where $\beta_i(\mathcal{G}) \equiv [\partial F_i(\mathcal{G}, z)/\partial z]_{z=1}$. The functions $\beta_i(\mathcal{G})$ may be calculated for small couplings in perturbation theory. This is Wilson's version of the "renormalization group" equation, which will be discussed in somewhat different terms in Volume II.

The Lagrangian for any finite value of the cutoff defines an effective field theory, in which instead of (or in addition to) integrating out "heavy" particles like the electron in the work of Euler *et al.*, one integrates out *all* particles with momenta greater than Λ. Even if one starts with a theory with a finite number of coupling parameters $\mathcal{G}_i^0$ at some cutoff Λ_0, at any other value of the cutoff the differential equation (12.4.3) will generally yield non-zero values for all couplings allowed by symmetry principles.[12]

We now distinguish between the renormalizable and unrenormalizable couplings, labelled $\mathcal{G}_a$ and $\mathcal{G}_n$, respectively, with a running over the finite number N of couplings (including masses) for which $\Delta_a \geq 0$, and n running over the infinite number of couplings with dimensionalities $\Delta_n < 0$. We want to show that if the couplings $\mathcal{G}_a(\Lambda_0)$ and $\mathcal{G}_n(\Lambda_0)$ at some initial cutoff value Λ_0 lie on a generic N-dimensional initial surface $\mathcal{S}_0$, then (with some qualifications) for $\Lambda \ll \Lambda_0$ they will approach a fixed surface $\mathcal{S}$ that is independent of both

[11] This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

[12] The only known exceptions to this rule are in theories based on supersymmetry [4].

Λ_0 and the initial surface.[13] This fixed surface is stable, in the sense that from any point on the surface, the trajectory generated by Eq. (12.4.3) stays on the surface. Such a stable surface defines a finite-parameter set of theories whose physical content is cutoff-independent, which as argued in the previous section, is the essential property of renormalizable theories. Furthermore, this construction shows that a generic theory defined with cutoff Λ_0 will look for $\Lambda \ll \Lambda_0$ like a renormalizable theory.[14]

To prove these results, consider any small perturbation $\delta\mathcal{G}_i(\Lambda)$ in the values of the $\mathcal{G}_i(\Lambda)$ satisfying Eq. (12.4.3). It will satisfy the differential equation

$$\Lambda \frac{d}{d\Lambda} \delta\mathcal{G}_i(\Lambda) = \sum_j M_{ij}\big(\mathcal{G}(\Lambda)\big)\delta\mathcal{G}_j(\Lambda), \tag{12.4.4}$$

where

$$M_{ij}(\mathcal{G}) \equiv \frac{\partial}{\partial \mathcal{G}_j}\beta_i(\mathcal{G}). \tag{12.4.5}$$

This equation couples the renormalizable and unrenormalizable couplings, making it difficult to see the difference in their behavior. To decouple them, we introduce the linear combinations

$$\xi_n \equiv \delta\mathcal{G}_n - \sum_{ab} \frac{\partial \mathcal{G}_n}{\partial \mathcal{G}^0_a}\left(\frac{\partial \mathcal{G}}{\partial \mathcal{G}^0}\right)^{-1}_{ab} \delta\mathcal{G}_b, \tag{12.4.6}$$

where $\mathcal{G}^0_a$ are the values of the renormalizable couplings at cutoff Λ_0, which we shall use as coordinates for the initial surface, and $\mathcal{G}_n$ are the values of the non-renormalizable couplings at cutoff Λ derived from the differential equation (12.4.3), with initial value for Λ_0 at the point on the initial surface with coordinates $\mathcal{G}^0_a$. To calculate the derivative of ξ_n with respect to Λ, we note that the derivatives $\partial\mathcal{G}_i/\partial\mathcal{G}^0_a$ satisfy the same differential equation (12.4.4) as the $\delta\mathcal{G}_i$. It is an elementary exercise then to show that

$$\Lambda \frac{d}{d\Lambda}\xi_n = \sum_m N_{nm}\xi_m, \tag{12.4.7}$$

where

$$N_{nm} \equiv M_{nm} - \sum_{ab} \frac{\partial \mathcal{G}_n}{\partial \mathcal{G}^0_a}\left(\frac{\partial \mathcal{G}}{\partial \mathcal{G}^0}\right)^{-1}_{ab} M_{bm}. \tag{12.4.8}$$

Now we must estimate the elements of N_{nm}. For a free-field theory no cutoff is needed, so for very small coupling all bare parameters $g_i(\Lambda)$ become Λ-independent. Hence for small couplings the dimensionless parameters $\mathcal{G}_i$ simply scale as $\Lambda^{-\Delta_i}$, and the matrix M_{ij} is given by

$$M_{ij} \approx -\Delta_i \delta_{ij}. \tag{12.4.9}$$

[13] This theorem is due to Polchinski [19]. What follows here is a shortened and less rigorous version. (In Polchinski's proof, the initial surface is taken to be that with all non-renormalizable couplings vanishing. As we shall see here, the couplings approach the same fixed surface for generic initial surfaces.)

[14] Of course some theories have symmetries and a field content that do not allow any renormalizable interactions. This is the case for theories containing only fermion fields, or only the gravitational field. Such theories for $\Lambda \ll \Lambda_0$ look like free-field theories.

It follows that the matrix N_{nm} is given approximately by $-\Delta_n\delta_{nm}$. The defining characteristic of the non-renormalizable couplings is that $\Delta_n < 0$, so Eq. (12.4.7) tells us that, at least for couplings in some finite range, where N_{nm} is positive-definite, the ξ_n decay for $\Lambda \ll \Lambda_0$ like positive powers of Λ/Λ_0. In this limit, then, the perturbations are related by

$$\delta\mathcal{G}_n = \sum_{ab} \frac{\partial \mathcal{G}_n}{\partial \mathcal{G}^0_a} \left(\frac{\partial \mathcal{G}}{\partial \mathcal{G}^0}\right)^{-1}_{ab} \delta\mathcal{G}_b. \tag{12.4.10}$$

In particular, if we make a small change in the initial surface $\mathcal{S}_0$ and/or the starting point on that surface and/or the initial cutoff Λ_0, such that the perturbations $\delta\mathcal{G}_a$ in the renormalizable couplings vanish at some cutoff $\Lambda \ll \Lambda_0$, then the perturbations $\delta\mathcal{G}_n$ in all the other couplings at cutoff Λ also vanish. Thus the non-renormalizable couplings $\mathcal{G}_n(\Lambda)$ for $\Lambda \ll \Lambda_0$ can depend only on the renormalizable couplings $\mathcal{G}_a(\Lambda)$, not separately on the initial surface or the starting point on that surface or the initial cutoff Λ_0. At cutoff $\Lambda \ll \Lambda_0$ all the couplings therefore approach an N-dimensional surface $\mathcal{S}$, with coordinates $\mathcal{G}_a(\Lambda)$, which is independent of both the initial surface and of Λ_0. Note that the non-renormalizable couplings $\mathcal{G}_n$ are not generally small on $\mathcal{S}$; the important point is that they become functions of the renormalizable couplings. Changes in Λ with Λ remaining much less than Λ_0 will change the couplings, but the couplings will remain close to $\mathcal{S}$ (at least as long as the couplings do not become so large that N_{nm} is no longer a positive-definite matrix). Hence $\mathcal{S}$ is a stable surface, as was to be proved.

We have seen that all physical quantities may be expressed in terms of Λ and the $\mathcal{G}_n(\Lambda)$, and are Λ-independent. This is true in particular of the N conventional renormalized couplings and masses, like e and m_e in quantum electrodynamics. But we can then invert this relation, and express the $\mathcal{G}_n(\Lambda)$ in terms of the conventional parameters and Λ. In this way we can justify the usual renormalization program; all physical quantities are expressed in a cutoff-independent way in terms of the conventional renormalized couplings and masses.

The Wilson approach has some advantages in practice. One does not have to worry about subintegrations and overlapping divergences; the momentum cutoff applies to all internal lines. Also, some of the non-renormalization theorems of supersymmetry theory, which tell us that certain couplings are not affected by radiative corrections, work only for the cutoff-dependent bare couplings [20].

On the other hand, there are disadvantages to the Wilson approach. One must give up the special simplicities of working with renormalizable theories like quantum electrodynamics; once one starts integrating out particles with momenta above some scale Λ, the resulting effective field theory will contain all Lorentz- and gauge-invariant interactions, with Λ-dependent couplings. (Nevertheless, in physical processes at energies $E \ll \Lambda$, the dominant couplings will still be the renormalizable ones.) Also, the cutoff generally destroys *manifest* gauge invariance, and either manifest Lorentz invariance or unitarity. None of this is a problem in condensed matter physics, the original context of Wilson's approach, because no one would expect a realistic condensed matter theory to be strictly renormalizable, and there are no fundamental physical principles that are necessarily violated by a cutoff. In fact, in crystals there is a cutoff on phonon momenta, provided by the inverse lattice spacing.

At bottom, the difference between the conventional and the Wilson approach is one of mathematical convenience rather than of physical interpretation. Indeed, conventional renormalization already provides a sort of adjustable cutoff; when we express our answer in terms of coupling constants that are defined as the values of physical amplitudes at some momenta of order μ (as for the scalar field theory discussed in the previous section), the cancellations that make integrals converge begin to operate at virtual momenta of order μ. Conversely, the Λ-dependent coupling constants of the Wilson approach must ultimately be expressed in terms of observable masses and charges, and when this is done the results are, of course, the same as those obtained by conventional means.

12.5 Accidental Symmetries[15]

In Section 12.3 we saw that there are good reasons to adopt renormalizable field theories as approximate descriptions of nature at sufficiently low energy. It often happens that condition of renormalizability is so stringent that the effective Lagrangian automatically obeys one or more symmetries, which are not symmetries of the underlying theory, and may therefore be violated by the suppressed non-renormalizable terms in the effective Lagrangian. Indeed, most of the experimentally discovered symmetries of elementary particle physics are "accidental symmetries" of this sort.

A classic example is provided by the inversions and flavor conservation in the electrodynamics of charged leptons. The most general renormalizable and gauge- and Lorentz-invariant Lagrangian density for photons and fields ψ_i of spin $\frac{1}{2}$ and charge $-e$ takes the form

$$\begin{aligned}\mathcal{L} = & -\frac{1}{4}Z_3 F_{\mu\nu}F^{\mu\nu} \\ & -\sum_{ij} Z_{Lij}\bar{\psi}_{Li}[\not{\partial} + ie\not{A}]\psi_{Lj} - \sum_{ij} Z_{Rij}\bar{\psi}_{Ri}[\not{\partial} + ie\not{A}]\psi_{Rj} \\ & -\sum_{ij} M_{ij}\bar{\psi}_{Li}\psi_{Rj} - \sum_{ij} M^\dagger_{ij}\bar{\psi}_{Ri}\psi_{Lj},\end{aligned} \tag{12.5.1}$$

where i, j are summed over the three lepton flavors (e, μ, and τ), ψ_{iL} and ψ_{iR} are the left- and right-handed parts of the field ψ_i, defined by

$$\psi_{Li} = \frac{1}{2}(\mathbf{1} + \gamma_5)\psi_i, \qquad \psi_{Ri} = \frac{1}{2}(\mathbf{1} - \gamma_5)\psi_i, \tag{12.5.2}$$

and Z_L, Z_R, and M are numerical matrices. We are not assuming anything about lepton flavor conservation, so the matrices Z_{Lij}, Z_{Rij} and M_{ij} need not be diagonal. Also we are not assuming anything about invariance under P, C, or T invariance, so there is no necessary relation between Z_L and Z_R, or between M and $M^\dagger$. The only constraints on these matrices come from the reality of the Lagrangian density, which requires that Z_{Lij} and Z_{Rij} are Hermitian, and from the canonical anticommutation relations, which require that Z_{Lij} and Z_{Rij} are positive-definite.

[15]This section lies somewhat out of the book's main line of development, and may be omitted in a first reading.

Now suppose we replace the lepton fields ψ_L, ψ_R with new fields ψ'_L, ψ'_R defined by

$$\psi_L = S_L \psi'_L, \qquad \psi_R = S_R \psi'_R, \tag{12.5.3}$$

where $S_{L,R}$ are non-singular matrices that can be chosen as we like. The Lagrangian density when expressed in terms of these new fields then takes the same form as in Eq. (12.5.1), but with new matrices

$$Z'_L = S_L^\dagger Z_L S_L, \qquad Z'_R = S_R^\dagger Z_R S_R, \qquad M' = S_L^\dagger M S_R. \tag{12.5.4}$$

We can choose S_L and S_R so that $Z'_L = Z'_R = \mathbf{1}$. (Take $S_{L,R} = U_{L,R} D_{L,R}$, where $U_{L,R}$ are the unitary matrices that diagonalize the positive-definite Hermitian matrices $Z_{L,R}$, and the $D_{L,R}$ are the diagonal matrices whose elements are the inverse square roots of the eigenvalues of $Z_{L,R}$.)

Now make another transformation, to lepton fields ψ''_L defined by

$$\psi'_L = S'_L \psi''_L, \qquad \psi'_R = S'_R \psi''_R. \tag{12.5.5}$$

The Lagrangian density again takes the same form when expressed in terms of these new fields, with new matrices

$$Z''_L = S'^\dagger_L S'_L, \qquad Z''_R = S'^\dagger_R S'_R, \qquad M'' = S'^\dagger_L M' S'_R. \tag{12.5.6}$$

This time we take $S'_{L,R}$ unitary, so that again $Z''_L = Z''_R = \mathbf{1}$. We choose these unitary matrices so that M'' is real and diagonal. (By the polar decomposition principle, M' like any square matrix may be put in the form $M' = VH$, where V is unitary and H is Hermitian. Take $S'_L = S'^\dagger_R V^\dagger$ and choose S'_R as the unitary matrix that diagonalizes H.) Dropping primes, the Lagrangian density now takes the form

$$\begin{aligned}\mathcal{L} = & -\frac{1}{4} Z_3 F_{\mu\nu} F^{\mu\nu} - \sum_i \bar{\psi}_{Li} [\not{\partial} + ie\not{A}] \psi_{Li} - \sum_i \bar{\psi}_{Ri} [\not{\partial} + ie\not{A}] \psi_{Ri} \\ & - \sum_i m_i \bar{\psi}_{Li} \psi_{Ri} - \sum_i m_i \bar{\psi}_{Ri} \psi_{Li},\end{aligned} \tag{12.5.7}$$

where m_i are real numbers, the eigenvalues of the Hermitian matrix H. Finally, this can be put in the more familiar form

$$\mathcal{L} = -\frac{1}{4} Z_3 F_{\mu\nu} F^{\mu\nu} - \sum_i \bar{\psi}_i [\not{\partial} + ie\not{A}] \psi_i - \sum_i m_i \bar{\psi}_i \psi_i. \tag{12.5.8}$$

With the Lagrangian taking this form, it is now apparent that any renormalizable Lagrangian for lepton electrodynamics automatically conserves P, C, and T, as well as the numbers of leptons (minus the numbers of antileptons) of each flavor: electron, muon, and tauon.[16] In particular, despite the appearance of Eq. (12.5.1), this theory does not allow

[16] This was first shown by Feinberg, Kabir, and myself [21]. Feinberg [22] had earlier noted that weak interaction effects in a theory with only one neutrino species would give rise to an observable rate for the process $\mu \to e + \gamma$, a difficulty that was only resolved by the discovery of a second neutrino species.

such processes as $\mu \to e+\gamma$. The reader may perhaps worry whether it is correct to identify the lepton fields as the ψ_i (previously called ψ''_i) appearing in Eq. (12.5.8), which obviously conserves lepton flavor, rather than the ψ_i appearing in Eq. (12.5.1), which seems to allow processes like $\mu \to e+\gamma$.

Such worries may be put aside; as stressed in Section 10.3, there is no one field that can be identified as *the* field of the electron or muon. In fact, although Eq. (12.5.1) yields a non-vanishing matrix element for radiative decay of lepton 1 into lepton 2 *off* the lepton mass shell, by taking the lepton momenta on the mass shell we find a vanishing S-matrix for all such processes even when calculated using Eq. (12.5.1).

It was essential in deriving these results that the same electric charge appeared in Eq. (12.5.1) for both the left- and right-handed parts of the lepton fields or, in other words, that both left- and right-handed parts of the lepton fields transform in the same way under electromagnetic gauge transformations. As we shall see in Volume II, for similar reasons the modern renormalizable theory of strong interactions known as quantum chromodynamics automatically conserves C, and (aside from certain non-perturbative effects) P and T, as well as the numbers of quarks (minus the numbers of antiquarks) of each quark flavor. We shall also see in Volume II that the simplest version of the renormalizable standard model of weak and electromagnetic interactions automatically conserves lepton flavor (though not C and P) for reasons similar to those described here for electrodynamics. It remains an open possibility that non-renormalizable interactions arising from higher mass scales may violate any of these conservation laws.

Problems

1. List all the renormalizable (or superrenormalizable) Lorentz-invariant terms in the Lagrangian of a single scalar field for space-time dimensionalities 2, 3, and 6.

2. Show how the overlapping divergence in the electron self-energy is cancelled in quantum electrodynamics.

3. Consider the theory of a scalar field ϕ and spinor field ψ, with interaction Hamiltonian $g\phi\bar{\psi}\psi$. Write the one-loop part of the scalar self-energy function $\Pi^*(q)$ as a divergent polynomial in p^μ, plus an explicit convergent integral.

4. Suppose that the quantum electrodynamics of electrons and photons is actually an effective field theory, derived by integrating out unknown particles of mass $M \gg m_e$. Assume gauge invariance and Lorentz invariance, but not invariance under C, P, or T. What are the non-renormalizable terms in the Lagrangian of leading order in $1/M$? Of next to leading order?

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